Circuit Lab, Division B

Practice Test 7 (National level, hard) · written test · 50 minutes · 50 questions

Team name: _____ Team number: _____

Names: _____

Instructions

- Write all answers on the answer sheet at the back. Only the answer sheet is graded.
- Give numeric answers in metric (SI) units with the correct number of significant figures.
- You may use two calculators and your own paper notes.
- Use $k = 8.99 \times 10^{9}\ \text{N·m}^2/\text{C}^2$ and $e = 1.602 \times 10^{-19}\ \text{C}$. Wires and meters are ideal unless the question says otherwise.
- In logic questions, A′ means NOT A, AB means A AND B, and A + B means A OR B.

Section	Questions
Section I: Multiple choice	1–15
Section II: Short answer and calculations	16–37
Section III: Circuit problems	38–50

Section I: Multiple choice

Circle the one best answer. 1 point each.

1. George Boole's FIRST book on logic came out in 1847, the same year as a public quarrel between Augustus De Morgan and Sir William Hamilton. What was it?
   - **A.** The Mathematical Analysis of Logic
   - **B.** An Investigation of the Laws of Thought
   - **C.** A Symbolic Analysis of Relay and Switching Circuits
   - **D.** Die galvanische Kette, mathematisch bearbeitet

2. In 1775 Volta improved and named a device that gives static charge again and again by induction, from one rubbing. It is the:
   - **A.** Leyden jar
   - **B.** Electrophorus
   - **C.** Torsion balance
   - **D.** Voltaic pile

3. In his electrical memoirs of the 1780s, Coulomb sampled charged conductors with a small "proof plane". He found that the charge on a conductor:
   - **A.** Fills its inside evenly
   - **B.** Collects at its center
   - **C.** Stays on its outer surface
   - **D.** Flows away at once through dry air

4. Hertz's spark-gap experiments (1886-1889) showed that electromagnetic waves exist and travel at the speed of light. This confirmed the theory of:
   - **A.** Isaac Newton
   - **B.** James Clerk Maxwell
   - **C.** Benjamin Franklin
   - **D.** Thomas Edison

5. Which statement about electric field lines is TRUE?
   - **A.** They form closed loops around a point charge
   - **B.** They begin on positive charges, end on negative charges, never cross, and are closer together where the field is stronger
   - **C.** They cross where two fields meet
   - **D.** They point from negative charges toward positive charges

6. Two point charges attract with a force F. One charge is doubled, and the distance between them is cut in half. The new force is:
   - **A.** F
   - **B.** 2F
   - **C.** 4F
   - **D.** 8F

7. Electricians follow a "one-hand rule" (one hand in a pocket) near live circuits. Why?
   - **A.** A shock then cannot take a path from hand to hand across the chest and heart
   - **B.** One hand has a higher resistance than two
   - **C.** It halves the voltage of the shock
   - **D.** It keeps the meter leads from touching

8. Which device makes a DC voltage directly from a TEMPERATURE DIFFERENCE between two junctions of different metals?

   A. Solar cell
   B. Thermocouple
   C. Piezoelectric crystal
   D. Fuel cell

9. A current of 2.5 mA flows through 4.20 kΩ. Written with the correct significant figures, the voltage is:

   A. 10.5 V
   B. 11 V
   C. 10.50 V
   D. 10 V

10. Three bulbs are in PARALLEL across a battery with a fuse. A wire is connected across ONE bulb. What happens?

    A. All three bulbs go out, because the wire shorts the battery and a very large current flows (the fuse blows)
    B. Only the shorted bulb goes out; the other two get brighter
    C. Only the shorted bulb goes out; the others do not change
    D. All three bulbs get brighter

11. In an SPDT relay, when is the NC (normally closed) contact connected to COM (common)?

    A. Only while the coil has current
    B. Only when the coil voltage is reversed
    C. Never; NC is a spare pin
    D. When the coil has no current

12. A 20 A breaker is put on a circuit wired with 14 AWG copper (rated for 15 A). What is the danger?

    A. The breaker will trip too often
    B. The voltage will be too low for appliances
    C. The wire can carry 15-20 A and overheat (a fire risk) without the breaker tripping
    D. There is no danger; a bigger breaker is always safer

13. A zener diode working in reverse breakdown:

    A. Keeps its voltage nearly constant while its current changes a lot
    B. Keeps its current constant while its voltage changes
    C. Blocks all current, like an open switch
    D. Gives off light whose color depends on the current

14. A full-wave bridge rectifier (four silicon diodes, 0.7 V each) is fed a sine wave with a peak of 12.0 V. About what is the peak voltage across the load?

    A. 12.0 V
    B. 11.3 V
    C. 10.6 V
    D. 8.5 V

15. Which set of gates is NOT enough to build every logic function (it is not functionally complete)?

    A. {AND, OR}
    B. {NAND}
    C. {NOR}
    D. {AND, NOT}

Section II: Short answer and calculations

Write the answer with units and the correct number of significant figures. 2 points each.

16. Four +1.0 μC charges sit at the corners of a square with sides of 0.10 m. Find the size and direction of the net force on one of them.

17. Charges +2.0 μC, −3.0 μC and +4.0 μC are placed at (0, 0), (0.30 m, 0) and (0, 0.40 m). Find the total electric potential energy of the group. Does it take positive work to bring them together from very far apart?

18. A metal sphere has a charge of −3.2 nC. It then LOSES 3.0×10^{10} electrons. Find its new charge.

19. Two identical cells (EMF 1.50 V, internal resistance 0.200 Ω each) drive a 1.00 Ω load. Find the load current when the cells are (a) in series and (b) in parallel.

20. A phone battery is 4000 mAh at 3.80 V. In standby the phone uses an average of 0.42 W. How long does a full battery last, in hours?

21. A town takes 1.00 MW through a line with a total resistance of 2.00 Ω. Find the power lost in the line if the power is sent at (a) 10.0 kV and (b) 100 kV.

22. A person with wet hands grabs a 120 V wire and a grounded pipe. The path through the body has 1.00 kΩ. Find the current. What can it do to the heart?

23. A 1.00 km reel of 12 AWG copper wire has a diameter of 2.05 mm ($\rho = 1.68 \times 10^{-8}$ Ω·m). Find its resistance and the voltage drop along it at 15.0 A.

24. A 12.0 V car battery is rated 55 Ah. How much charge (in coulombs) and how much energy (in joules) does it store?

25. (a) Two SMD resistors are marked 1003 and 220. Give their values. (b) Give the 5 color bands for a 3.30 kΩ, ±0.5% resistor.

26. To measure a resistor, a voltmeter is put across it and an ammeter in series with the pair. The voltmeter (internal resistance 10.0 kΩ) reads 4.80 V and the ammeter reads 2.40 mA. Find the TRUE resistance.

27. Twelve 10.0 Ω resistors form a 2-by-2 square grid (3 × 3 = 9 junctions, like a tic-tac-toe board with its outside edges). Find the resistance between two OPPOSITE corners.

28. A 5.60 kW clothes dryer runs on 240 V. It is a continuous load, so the breaker must be loaded to at most 80% of its rating. Find the dryer current and the smallest standard breaker (20, 25, 30 or 40 A).

29. A 12.0 V car circuit feeds two 55 W headlights and one 21 W lamp in parallel. The fuse rating must be at least 125% of the load current. Find the load current and choose the fuse (10, 15, 20 or 25 A).

30. A 5.60 V zener regulator must hold a 280 Ω load at 5.60 V while the supply varies from 9.00 V to 12.00 V. (a) What is the LARGEST series resistor that works? (b) With that resistor, what power does the zener use when the supply is 12.00 V?

31. Use a K-map to find the simplest sum of products for F(A, B, C, D) = Σm(0, 2, 5, 7, 8, 10, 13) with a don't care d(15). Which single 2-input gate does F match?

32. Add the THREE binary numbers 10110 + 11101 + 1111. Give the sum in binary and in decimal.

33. An 8-bit ripple-carry adder (eight full adders, bit 0 = the right-most bit) adds A = 10110101 and B = 01101011 with carry-in 0. (a) What is the carry INTO the full adder for bit 4? (b) Give the 8-bit sum S and the carry-out.

34. F(A, B, C, D) = Σm(1, 2, 3, 6, 9, 12, 13, 14) is built with ONE 8-to-1 multiplexer. A, B, C are the select inputs (A is the high bit). Give the data inputs D_0 to D_7 (each 0, 1, D or D′).

35. A USB-C charger label reads: 5 V ⎓ 3 A, 9 V ⎓ 3 A, 15 V ⎓ 3 A, 20 V ⎓ 3.25 A. (a) Give the power of each level. (b) A laptop's 57.0 Wh battery charges from 20% to 100% at the highest level, with 85.0% efficiency. Estimate the time in minutes.

36. Equal-current task: a red LED (V_F = 1.80 V) and a blue LED (V_F = 3.00 V) must carry the SAME current, 15.0 mA, from a 9.00 V battery, using only ONE resistor. How are they wired, and what resistor is needed?

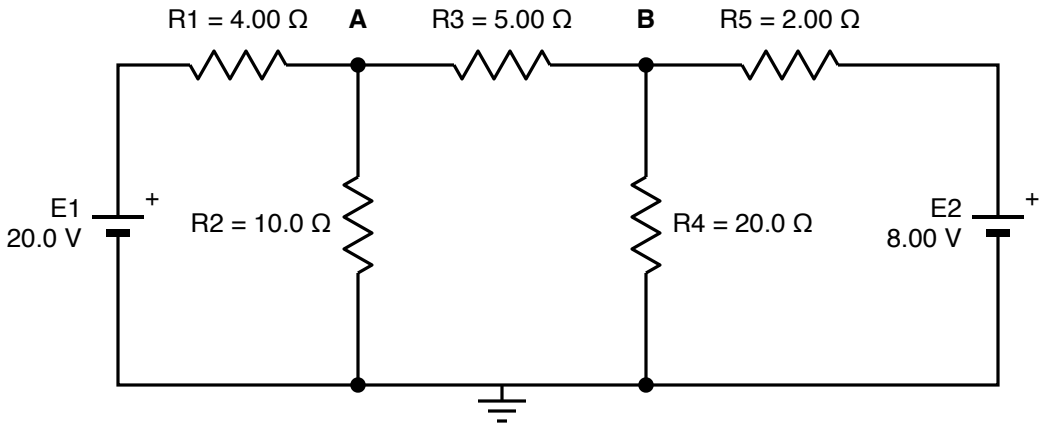
37. You have one each of 150 Ω, 220 Ω, 330 Ω, 470 Ω and 1.00 kΩ. Using at most four of them, make a resistance as close as possible to 290 Ω. Show the arrangement and its value.

Section III: Circuit problems

Show your work. 2 points per part.

Problem A: two batteries, two nodes (Kirchhoff's laws) (Questions 38–41)

E1 = 20.0 V feeds node A through R1 = 4.00 Ω. R2 = 10.0 Ω joins A to the bottom wire (ground). R3 = 5.00 Ω joins A to node B. R4 = 20.0 Ω joins B to ground. E2 = 8.00 V connects to B through R5 = 2.00 Ω. Both batteries have their + plate at the TOP. Let V_A and V_B be the node voltages above ground.



38. Write KCL at A and at B (two equations in V_A and V_B) and find V_A.

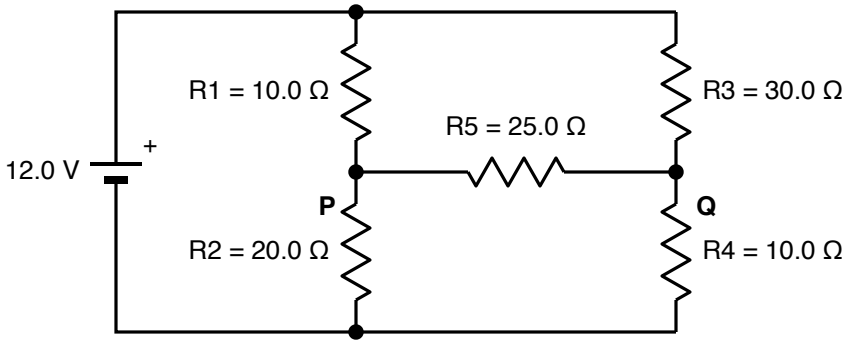
39. Find V_B.

40. Find the current in R3 and its direction.

41. Is E2 giving power to the circuit or being charged? Find the current in E2 and that power.

Problem B: a bridge that is not balanced (Questions 42–45)

A 12.0 V ideal battery feeds the bridge below. P is between R1 = 10.0 Ω (top) and R2 = 20.0 Ω (bottom). Q is between R3 = 30.0 Ω (top) and R4 = 10.0 Ω (bottom). R5 = 25.0 Ω joins P and Q.



42. First, R5 is removed. Find $V_P - V_Q$.

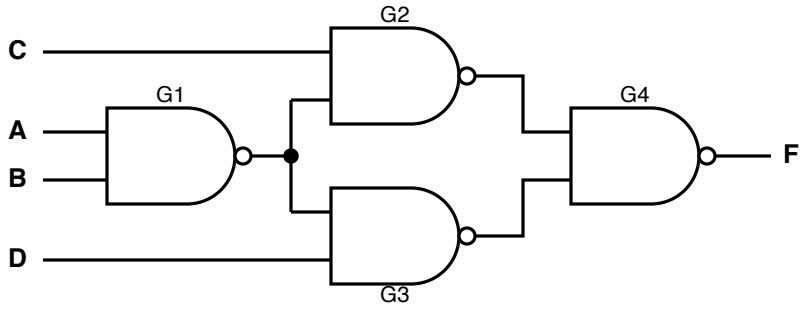
43. With R5 in place, write the node equations for P and Q and find the current in R5 and its direction.

44. With R5 in place, find the equivalent resistance that the battery sees.

45. Instead, R5 is replaced by a plain wire (P and Q are shorted together). Find the battery current.

Problem C: NAND-only logic (Questions 46–48)

Four NAND gates: G1 = NAND(A, B), G2 = NAND(C, G1), G3 = NAND(G1, D), and F = G4 = NAND(G2, G3).



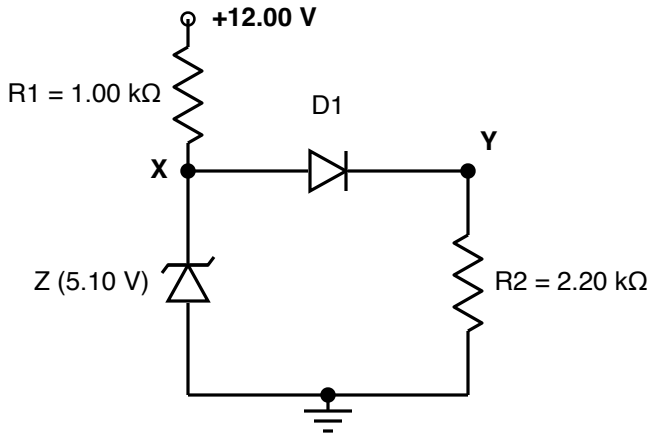
46. Write F and simplify it to the shortest sum of products.

47. For how many of the 16 input rows is F = 1? List them as minterm numbers.

48. Write F as a product of two sums (POS form).

Problem D: zener and diode (Questions 49–50)

+12.00 V feeds R1 = 1.00 kΩ to node X. A 5.10 V zener diode goes from X to ground (cathode at X). A silicon diode D1 (0.70 V when on, anode at X) goes from X to node Y. R2 = 2.20 kΩ joins Y to ground.



49. Find V_X and V_Y. Show that the zener is in breakdown.

50. Find the zener current and the power the zener uses.

End of the test. Check your answer sheet.

Answer sheet · Practice Test 7 (National level, hard)

Team name: _____ Team number: _____

Multiple choice: circle one letter. Other questions: write the answer with units.

1. A B C D
2. A B C D
3. A B C D
4. A B C D
5. A B C D
6. A B C D
7. A B C D
8. A B C D
9. A B C D
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